

$$\text{let } A = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix}$$

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$T(u) = Au$$

$$T(u+v) = A(u+v)$$

$$A(u+v) = Au + Av$$

$$\therefore T(u+v) = T(u) + T(v)$$

$$T(cu) = A(cu)$$

$$A(cu) = c(Au)$$

$$\therefore T(cu) = cT(u)$$

$$u = (2, -1)$$

$$u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \times \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix}$$

$$T(2, -1) = (6, 3, 0)$$

$$\text{let } A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

by $T(u) = Au$, For every $u \in \mathbb{R}^2$
Find $T\left(\begin{bmatrix} 1 \\ 4 \end{bmatrix}\right)$

$$T\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 0 \\ 0 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \checkmark$$

$$\forall u, v \in \mathbb{R}^2, c \in \mathbb{R}$$

$$[1] \quad T(u+v) = A(u+v) = Au + Av$$

$$= T(u) + T(v)$$

يمكن اطلاق القابلية بـ (1) $\checkmark$

$$[2] \quad T(cu) = A(cu) = c(Au)$$

$$= c T(u) \rightarrow (2) \quad \checkmark$$

$\therefore T$ is a Linear Transformation

$$T\left(\begin{bmatrix} 1 \\ 4 \end{bmatrix}\right) = A \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}_{2 \times 2} \begin{pmatrix} 1 \\ 4 \end{pmatrix}_{2 \times 1} = \begin{pmatrix} 9 \\ 3 \end{pmatrix}$$



$$R_3 = -2R_1 + R_3$$

$$R_4 = -R_1 + R_4$$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -5 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & -6 & -6 & -1 & -19 \end{array} \right]$$

$$\begin{array}{rrrrr} -2 & -4 & 2 & 0 & -4 \\ 2 & 4 & 1 & -3 & -2 \\ \hline 0 & 0 & 3 & -3 & -6 \end{array}$$

$$R_1 = -2R_2 + R_1$$

$$R_4 = 6R_2 + R_4$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 4 & 8 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & 0 & 0 & -13 & -39 \end{array} \right]$$

$$R_3 = \frac{1}{3}R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 4 & 8 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & -13 & -39 \end{array} \right]$$

$$R_1 = 3R_3 + R_1$$

$$R_2 = -R_3 + R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & -13 & -39 \end{array} \right]$$

$$\text{sol. } x = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$$

$$R_4 = -\frac{1}{13}R_4$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 1 & 3 \end{array} \right]$$

$$R_1 = -R_4 + R_1$$

$$R_2 = R_4 + R_2$$

$$R_3 = R_4 + R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \end{array} \right]$$

$$x_1 = -1$$

$$x_2 = 2$$

$$x_3 = 1$$

$$x_4 = 3$$

ref

$$\begin{aligned}x + y + 2z - 5w &= 3 \\2x + 5y - z - 9w &= -3 \\2x + y - z + 3w &= -11 \\x - 3y + 2z + 7w &= -5\end{aligned}$$

Part (7)

$$\begin{bmatrix} \textcircled{1} & 1 & 2 & -5 & 3 \\ 2 & 5 & -1 & -9 & -3 \\ 2 & 1 & -1 & 3 & -11 \\ 1 & -3 & 2 & 7 & -5 \end{bmatrix} \quad \begin{aligned} R_2 &= -2R_1 + R_2 \\ R_3 &= -2R_1 + R_3 \\ R_4 &= -R_1 + R_4 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & 2 & -5 & 3 \\ 0 & 3 & -5 & 1 & -9 \\ 0 & -2 & -5 & 13 & -17 \\ 0 & -4 & 0 & 12 & -8 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 1 & 2 & -5 & 3 \\ 0 & -2 & -5 & 13 & -17 \\ 0 & 3 & -5 & 1 & -9 \\ 0 & -4 & 0 & 12 & -8 \end{bmatrix}$$

$$\xrightarrow{R_2 = -R_2} \begin{bmatrix} 1 & 1 & 2 & -5 & 3 \\ 0 & 1 & 5 & -13 & 17 \\ 0 & 3 & -5 & 1 & -9 \\ 0 & -4 & 0 & 12 & -8 \end{bmatrix} \quad \begin{aligned} R_1 &= -R_2 + R_1 \\ R_3 &= -3R_2 + R_3 \\ R_4 &= 4R_2 + R_4 \end{aligned} \quad \begin{bmatrix} 1 & 0 & -3 & 8 & -14 \\ 0 & 1 & 5 & -13 & 17 \\ 0 & 0 & -20 & 40 & -60 \\ 0 & 0 & 20 & -40 & 60 \end{bmatrix}$$

$$\xrightarrow{R_3 = \frac{-1}{20}R_3} \begin{bmatrix} 1 & 0 & -3 & 8 & -14 \\ 0 & 1 & 5 & -13 & 17 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 20 & -40 & 60 \end{bmatrix} \quad \begin{aligned} R_1 &= 3R_3 + R_1 \\ R_2 &= -5R_3 + R_2 \\ R_4 &= -20R_3 + R_4 \end{aligned} \quad \begin{bmatrix} \textcircled{1} & x & y & z & w \\ 1 & 0 & 0 & 2 & -5 \\ 0 & 1 & 0 & -3 & 2 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Let $w=t, t \in \mathbb{R}$

$$z - 2w = 3$$

$$z - 2t = 3$$

$$z = 2t + 3$$

$$x + 2w = -5$$

$$x + 2t = -5$$

$$x = -2t - 5$$

$$y - 3w = 2$$

$$y - 3t = 2$$

$$y = 3t + 2$$

Answer

$$\text{soln. } X = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -2t-5 \\ 3t+2 \\ 2t+3 \\ t \end{pmatrix} \quad \checkmark t \in \mathbb{R}$$

$$\left(\begin{array}{cccc|c} x & y & z & w & \\ \hline \textcircled{1} & 0 & 0 & 2 & 3 \\ 0 & \textcircled{2} & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} \rightarrow x+2w=3 \\ \rightarrow x+2y=3 \\ \boxed{x=-2y+3} \\ y+z=1 \\ y+t=2 \rightarrow \boxed{y=-t+1} \end{array}$$

Let $z=t$ & $w=r$, $t, r \in \mathbb{R}$

$$\text{soln. } X = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -2r+3 \\ -t+1 \\ t \\ r \end{pmatrix} \quad \checkmark t, r \in \mathbb{R}$$

Part 5

$$\begin{aligned} 3x + 3y + 12z &= 6 \\ x + y + 4z &= 2 \\ 2x + 5y + 20z &= 10 \\ -x + 2y + 8z &= 4 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 3 & 3 & 12 & 6 \\ 1 & 1 & 4 & 2 \\ 2 & 5 & 20 & 10 \\ -1 & 2 & 8 & 4 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 4 & 2 \\ 3 & 3 & 12 & 6 \\ 2 & 5 & 20 & 10 \\ -1 & 2 & 8 & 4 \end{array} \right]$$

$$\begin{aligned} R_2 &= -3R_1 + R_2 \\ R_3 &= -2R_1 + R_3 \end{aligned}$$

$$R_4 = R_1 + R_4$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 3 & 14 & 6 \\ 0 & 3 & 12 & 6 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_4}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 4 & 2 \\ 0 & 3 & 12 & 0 \\ 0 & 3 & 14 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_2 = \frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 4 & 2 \\ 0 & 1 & 4 & 2 \\ 0 & 3 & 14 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{aligned} R_1 &= -R_2 + R_1 \\ R_3 &= -3R_2 + R_3 \end{aligned} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{aligned} &\text{refF} \\ &\text{let } z = t, t \in \mathbb{R} \end{aligned}$$

$$\text{Solu. } x = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -4t+2 \\ t \end{pmatrix} \quad \begin{aligned} &y+4z=2 \\ &y+4t=2 \\ &y=-4t+2 \end{aligned}$$

infinit mini solutions

$$x + y = 2$$

$$y + z = 2$$

$$x + z = 2$$

$$x + y - 2z = 0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \end{array} \right]$$

$$\begin{array}{l} R_3 = -R_1 + R_3 \\ R_4 = -R_1 + R_4 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -2 & -2 \end{array} \right]$$

$$\begin{array}{l} R_1 = -R_2 + R_1 \\ R_3 = R_2 + R_3 \end{array} \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -2 & -2 \end{array} \right)$$

$$R_3 = \frac{1}{2} R_3 \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -2 & -2 \end{array} \right)$$

$$\begin{array}{l} R_1 = R_3 + R_1 \\ R_2 = -R_3 + R_2 \\ R_4 = 2R_3 + R_4 \end{array} \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

rref

$$\boxed{x = 1}$$

$$\boxed{y = 1}$$

$$\boxed{z = 1}$$

unique solutions

$\therefore$ # of leading 1's =

of variables =

$\therefore$ this a unique solution

$$\text{Soln. } X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Part 6

$$\begin{aligned} x + 2y + 3z + 4w &= 5 \\ x + 3y + 5z + 7w &= 11 \\ x &\quad - z - 2w = -6 \end{aligned}$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 5 & 7 & 11 \\ 1 & 0 & -1 & -2 & -6 \end{array} \right) \xrightarrow{\substack{R_2 = -R_1 + R_2 \\ R_3 = -R_1 + R_3}} \left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 6 \\ 0 & -2 & -4 & -6 & -11 \end{array} \right)$$

$$\begin{aligned} R_1 &= -2R_2 + R_1 \\ R_3 &= 2R_2 + R_3 \end{aligned} \quad \left(\begin{array}{cccc|c} 1 & 0 & -1 & -2 & -1 \\ 0 & 1 & 2 & 3 & 6 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right)$$

$\therefore 0 = 1 \quad \therefore \text{No solution}$ ← صفر يساوي الرقم

$$\underline{R_2 = -R_2} \rightarrow \left[\begin{array}{cccc|c} \textcircled{1} & 1 & 1 & 1 & 0 \\ 0 & \textcircled{1} & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \end{array} \right] \begin{array}{l} R_1 = -R_2 + R_1 \\ R_3 = -R_2 + R_3 \end{array}$$

$$\left[\begin{array}{cccc|c} \textcircled{1} & 0 & 0 & 1 & 0 \\ 0 & \textcircled{1} & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \end{array} \right] \underline{R_3 = -R_3}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right] \underline{R_2 = -R_3 + R_2}$$

$$\left[\begin{array}{cccc|c} x & y & z & w & \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right] \begin{array}{l} \text{rref} \\ \rightarrow \begin{array}{l} x + w = 0 \\ x + t = 0 \rightarrow \boxed{x = -t} \\ z + w = 0 \\ z + t = 0 \end{array} \\ \rightarrow \boxed{z = -t} \end{array}$$

Let $w = t, t \in \mathbb{R}$

$$y - w = 0$$

$$y - t = 0$$

$$\boxed{y = t}$$

$$\therefore \text{Soln. } X \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -t \\ t \\ -t \\ t \end{pmatrix}, t \in \mathbb{R}$$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & -4 & 3 & 6 & 0 & 1 \end{array} \right]$$

$$R_2 = R_2 + R_1$$

$$R_3 = 4R_2 + R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 4 & 1 \end{array} \right]$$

$$R_3 = -R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{array} \right]$$

$$R_1 = R_3 + R_1$$

$$R_2 = R_3 + R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -3 & -1 \\ 0 & 1 & 0 & -3 & -3 & -1 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{array} \right]$$

I

A⁻¹

Part 18

Evaluate

$$\begin{vmatrix} + & 2 & 3 & 5 & 7 \\ - & 0 & 0 & 3 & 4 \\ + & 0 & 4 & 1 & 0 \\ - & 0 & 0 & 0 & -1 \end{vmatrix}$$

$$= 2 \begin{vmatrix} + & 0 & 3 & 4 \\ - & 4 & 1 & 0 \\ + & 0 & 0 & -1 \end{vmatrix} = 2 \begin{bmatrix} -4 & 3 & 4 \\ 0 & -1 \end{bmatrix}$$

$$2 [-4(3(-1))]$$

$$2 \times 12 = 24$$

$$\begin{vmatrix} 2 & 3 & 5 & 7 \\ 0 & 0 & 3 & 4 \\ 0 & 4 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix} = - \begin{vmatrix} 2 & 3 & 5 & 7 \\ 0 & 4 & 1 & 0 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & -1 \end{vmatrix}$$

$$-2 \times 4 \times 3 \times -1 = 24$$

$$R_2 \leftrightarrow R_3$$



By reduction to triangular form, evaluate

$$\begin{vmatrix} 1 & -3 & -4 & -2 \\ -1 & 4 & 3 & 4 \\ 2 & -5 & 1 & 6 \\ 0 & 1 & 0 & -1 \end{vmatrix}$$

$$\begin{aligned} R_2 &= R_1 + R_2 \\ R_3 &= 2R_1 + R_3 \end{aligned}$$

$$\begin{vmatrix} 1 & -3 & -4 & -2 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & 9 & 12 \\ 0 & 1 & 0 & -1 \end{vmatrix}$$

$$R_3 = -R_2 + R_3$$

$$R_4 = -R_2 + R_4$$

$$\begin{vmatrix} 1 & -3 & -4 & -2 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 10 & 10 \\ 0 & 0 & 1 & -3 \end{vmatrix}$$

10 common factor
from row 3

$$10 \begin{vmatrix} 1 & -3 & -4 & -2 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -3 \end{vmatrix}$$

$$R_4 = -R_3 + R_4$$

$$= 10 \begin{vmatrix} 1 & -3 & -4 & -2 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -4 \end{vmatrix}$$

$$= 10 \times 1 \times 1 \times 1 \times -4 = -40$$

$$|A_2| = \begin{vmatrix} 4 & 10 \\ 3 & 11 \end{vmatrix} = (4)(11) - (3)(10) = 44 - 30 = 14$$

$$x_1 = \frac{|A_1|}{|A|} = \frac{-28}{-14} = 2$$

$$x_2 = \frac{|A_2|}{|A|} = \frac{14}{-14} = -1$$

$$\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

$$|A| = \begin{vmatrix} 1 & -2 & 3 \\ -1 & 3 & 0 \\ 2 & -5 & 5 \end{vmatrix} = 1 \begin{vmatrix} -2 & 3 \\ -5 & 5 \end{vmatrix} + 3 \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix}$$

$$x = \frac{|A_1|}{|A|} = \frac{2}{2} = 1 \quad = [(-2)(5) - (5)(3)] + 3[(1)(5) - (2)(5)]$$

$$y = \frac{|A_2|}{|A|} = \frac{-2}{2} = -1 \quad \text{Soln}$$

$$z = \frac{|A_3|}{|A|} = \frac{4}{2} = 2 \quad = [-10 + 15] + 3[5 - 0] = 5 - 3 = 2$$

$$|A_1| = \begin{vmatrix} 9 & -2 & 3 \\ -4 & 3 & 0 \\ 17 & -5 & 5 \end{vmatrix} = 9 \begin{vmatrix} -2 & 3 \\ -5 & 5 \end{vmatrix} + 3 \begin{vmatrix} 9 & 3 \\ 17 & 5 \end{vmatrix} = 4 \begin{vmatrix} -2 & 3 \\ -5 & 5 \end{vmatrix} + 3 \begin{vmatrix} 9 & 3 \\ 17 & 5 \end{vmatrix} = 4[-2 \times 5 - (-5 \times 3)] + 3[9 \times 5 - 17 \times 3] = 20 - 18 = 2$$

$$|A_2| = \begin{vmatrix} 1 & 9 & 3 \\ -1 & -4 & 0 \\ 2 & 17 & 5 \end{vmatrix} = -1 \begin{vmatrix} 9 & 3 \\ 17 & 5 \end{vmatrix} + 4 \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = -6 - 4(-1) = -2$$

$$|A_3| = \begin{vmatrix} 1 & 2 & 3 \\ -1 & 3 & 0 \\ 2 & -5 & 5 \end{vmatrix} = 1 \begin{vmatrix} 2 & 3 \\ -1 & 5 \end{vmatrix} + 3 \begin{vmatrix} 1 & 9 \\ 2 & 17 \end{vmatrix} + 4 \begin{vmatrix} 1 & -2 \\ 2 & -5 \end{vmatrix} = 4$$

$$2f \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 2, \text{ Find } \begin{vmatrix} 2g & a & 3d \\ 2h & b & 3e \\ 2i & c & 3f \end{vmatrix}$$

$$\begin{vmatrix} 2g & a & 3d \\ 2h & b & 3e \\ 2i & c & 3f \end{vmatrix} \xrightarrow{C_1 \leftrightarrow C_2} - \begin{vmatrix} a & 2g & 3d \\ b & 2h & 3e \\ c & 2i & 3f \end{vmatrix}$$

$$\xrightarrow{C_2 \leftrightarrow C_3} = (-)(-)(-) \begin{vmatrix} a & 3d & 2g \\ b & 3e & 2h \\ c & 3f & 2i \end{vmatrix}$$

$$= 3 \begin{vmatrix} a & d & 2g \\ b & e & 2h \\ c & f & 2i \end{vmatrix}$$

$$= (3)(2) \begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix}$$

$$\det(A) = \det(A^T)$$

$$\textcircled{T} \begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = (3)(2)(2) = 12$$

$$M_{23} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

$$M_{31} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1$$

$$M_{32} = \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1$$

$$M_{33} = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1$$

$$C_{ij} = (-1)^{i+j} (M_{ij})$$

(2) Compute cofactor matrix C

$$C = \begin{pmatrix} +2 & -3 & +1 \\ -2 & +2 & -1 \\ +1 & -1 & +1 \end{pmatrix} = \begin{pmatrix} 2 & -3 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

(3) $\text{adj}(A) = C^T$:

$$\text{adj}(A) = \begin{pmatrix} 2 & -2 & 1 \\ -3 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

$$A = \begin{bmatrix} + & - & + & - \\ 1 & 2 & 3 & 0 \\ - & + & - & + \\ 1 & 1 & 0 & 2 \\ + & - & + & - \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}$$

$$\cancel{|A| = 2 \begin{vmatrix} + & - & + \\ 1 & 3 & 0 \\ 1 & 0 & 2 \\ 3 & 0 & -2 \end{vmatrix} - 3 \begin{vmatrix} + & - & + \\ 1 & 2 & 3 \\ 1 & 1 & 0 \\ 3 & 4 & 0 \end{vmatrix}}$$

$$\cancel{\begin{vmatrix} 0 & 2 & -3 \\ 0 & -2 & -3 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix} + \begin{vmatrix} 2 & 3 \\ 4 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 3 & 0 \end{vmatrix}}$$

$$\cancel{0 + 24}$$

$$\cancel{2 \times 48}$$

$$\cancel{-12 + -9}$$

$$\cancel{= -30 \times 3 = -90}$$

$$\cancel{49 - 90 =}$$

$$-3 \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix}$$

$$= 3 \times 13 = 39$$

If the set $M_{2 \times 2}$ of all 2×2 matrices with entries in $\mathbb{R}$ forms a vector space over $\mathbb{R}$, prove that W is a subspace of $M_{2 \times 2}(\mathbb{R})$, where

$$W = \left\{ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & 0 \end{bmatrix} : a_{11}, a_{12}, a_{21} \in \mathbb{R} \right\}$$

$$\text{Let } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & 0 \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & 0 \end{pmatrix}$$

$$A+B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & 0 \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & 0 \end{pmatrix} = \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & 0 \end{pmatrix} \in W$$

show that $W = \left\{ \begin{bmatrix} a \\ b \\ a+b \end{bmatrix} : a, b \in \mathbb{R} \right\}$ is a subspace of the vector space $\mathbb{R}^3$

$$U = \begin{pmatrix} a_1 \\ b_1 \\ a_1+b_1 \end{pmatrix}, V = \begin{pmatrix} a_2 \\ b_2 \\ a_2+b_2 \end{pmatrix}$$

$$\textcircled{1} U+V = \begin{pmatrix} a_1 \\ b_1 \\ a_1+b_1 \end{pmatrix} + \begin{pmatrix} a_2 \\ b_2 \\ a_2+b_2 \end{pmatrix} = \begin{pmatrix} a_1+a_2 \\ b_1+b_2 \\ a_1+a_2+b_1+b_2 \end{pmatrix}$$

$$\textcircled{2} \alpha U = \alpha \begin{pmatrix} a_1 \\ b_1 \\ a_1+b_1 \end{pmatrix} = \begin{pmatrix} \alpha a_1 \\ \alpha b_1 \\ \alpha a_1 + \alpha b_1 \end{pmatrix}$$

Math 2
vector 9/15

$$C_1 v_1 + C_2 v_2$$

$$\begin{pmatrix} 9 \\ 2 \\ 7 \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + C_2 \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} C_1 \\ 2C_1 \\ -C_1 \end{pmatrix} + \begin{pmatrix} 6C_2 \\ 4C_2 \\ 2C_2 \end{pmatrix}$$

$$\begin{pmatrix} 9 \\ 2 \\ 7 \end{pmatrix} = \begin{pmatrix} C_1 + 6C_2 \\ 2C_1 + 4C_2 \\ -C_1 + 2C_2 \end{pmatrix}$$

$$C_1 + 6C_2 = 9$$

$$2C_1 + 4C_2 = 2$$

$$-C_1 + 2C_2 = 7$$

$$C_1 = -3$$

$$C_2 = 2$$

$$2V_1 + 3V_2 - 5V_3 = 0$$

$$2V_1 + 3V_2 = 5V_3$$

$V_1 = (1, 1, 0)$, $V_2 = (5, 1, 1)$, $V_3 = (2, 7, 4)$
in R_3 from a linearly dependent or independent?

soln.

$$C_1 V_1 + C_2 V_2 + C_3 V_3 = (0, 0, 0)$$

$$C_1(1, 1, 0) + C_2(5, 1, 1) + C_3(2, 7, 4) = (0, 0, 0)$$

$$C_1 + 5C_2 + 2C_3 = 0$$

$$C_1 + C_2 + 7C_3 = 0$$

$$C_2 + 4C_3 = 0$$

$$\left| \begin{array}{ccc|c} 1 & 5 & 2 & 0 \\ 1 & 1 & 7 & 0 \\ 0 & 1 & 4 & 0 \end{array} \right| = (-3) - 18 = -21$$

span independent
Basis

$$\Delta \neq 0 \leadsto \text{independent}$$

$$\Delta = 0 \leadsto \text{dependent}$$

نظام غير متجانس

$$AX = b$$

$$x = A^{-1}b$$

$$AX = 0$$

نظام متجانس

Find the eigenvalues of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and the corresponding eigen vectors

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = 0$$

$$(2-\lambda)^2 - 1 = 0 \longrightarrow 4 - 4\lambda + \lambda^2 - 1 = 0$$

$$\boxed{\lambda^2 - 4\lambda + 3 = 0}$$

$$(\lambda - 1)(\lambda - 3) = 0$$

$$\lambda = 1$$

$$\lambda = 3$$

$$\left. \begin{array}{l} \lambda = 1 \quad \begin{pmatrix} 2-1 & 1 \\ 1 & 2-1 \end{pmatrix} \quad \text{let } X = \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ x + y = 0 \\ x = -y \end{array} \right\} \lambda = 3 \quad \begin{pmatrix} 2-3 & 1 \\ 1 & 2-3 \end{pmatrix} \quad \text{let } X = \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ -x + y = 0 \\ x - y = 0 \quad x = y \end{array}$$

$$\begin{pmatrix} x \\ -x \end{pmatrix} = x \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ x \end{pmatrix} \rightarrow x \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{vmatrix} 3-\lambda & 0 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$(3-\lambda) \times (2-\lambda) = 0$$

$$\lambda = 2, \lambda = 3$$

$$\begin{pmatrix} 3-\lambda & 0 \\ 1 & 2-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{at } \lambda = 2 \quad \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x + 0y = 0$$

$$\begin{pmatrix} 0 \\ y \end{pmatrix} = y \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{bmatrix} 3 & 1 & 0 \\ -4 & -1 & 0 \\ 4 & -8 & -2 \end{bmatrix}$$

$$C = \begin{vmatrix} -1 & 0 \\ -8 & -2 \end{vmatrix} + \begin{vmatrix} 3 & 0 \\ 4 & -2 \end{vmatrix} + \begin{vmatrix} 3 & 1 \\ -4 & -1 \end{vmatrix} = 2 - 6 + 1 = -3$$

$$\lambda^3 - 0\lambda^2 - 3\lambda - 2 = 0$$

$$A^3 - 3A + 2I = 0$$

$$A^{-1}A^3 - 3A^{-1}A + 2A^{-1}I = 0$$

$$A^2 - 3I + 2A^{-1} = 0$$

$$A^{-1} = -\frac{1}{2} [A^2 - 3I]$$

$$|A| = \lambda_1 \lambda_2 \dots \lambda_n$$

اضرب عناصر القطر الرئيس

SCALAR multiplication

The product of a vector

Exp

let $u = \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix}$ and $v = \begin{bmatrix} -2 \\ 2 \\ -3 \end{bmatrix}$ be vectors in $\mathbb{R}^3$

Find
 $3u - 2v$

$$3 \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix} - 2 \begin{bmatrix} -2 \\ 2 \\ -3 \end{bmatrix} =$$

$$\begin{bmatrix} 3 \\ 0 \\ 15 \end{bmatrix} + \begin{bmatrix} 4 \\ -4 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ -4 \\ 21 \end{bmatrix}$$

$$V = \begin{bmatrix} 10 \\ 1 \\ 4 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \text{ and } u_3 = \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 10 \\ 1 \\ 4 \end{bmatrix} = C_1 \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + C_3 \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix}$$

$$2C_1 + C_2 - 2C_3 = 10$$

$$3C_1 + 2C_2 + 2C_3 = 1$$

$$5C_1 + 4C_2 + 3C_3 = 4$$

$$C_1 = 1$$

$$C_2 = 2$$

$$C_3 = -3$$

$$V = C_1 u_1 + C_2 u_2 + C_3 u_3$$

$$V = u_1 + 2C_2 - 3u_3$$

Determine whether the vector $X = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$ is a linear combination of the vectors

$$X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ and } X_2 = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

$$X = aX_1 + bX_2$$

$$\therefore X = 2X_1 + X_2$$

$\therefore X$ Linear combination of X_1 and X_2

$$\begin{bmatrix} 5 \\ 0 \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ 2a \end{bmatrix} + \begin{bmatrix} 3b \\ -4b \end{bmatrix} = \begin{bmatrix} a + 3b \\ 2a - 4b \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 2 & -4 & 0 \end{array} \right] \xrightarrow{R_2 = -2R_1 + R_2} \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -10 & -10 \end{array} \right]$$

$$\xrightarrow{R_2 = \frac{1}{-10}R_2} \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & 1 & 1 \end{array} \right] \xrightarrow{R_1 = -3R_2 + R_1} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right] \xrightarrow{\text{RREF}}$$

$$\therefore a = 2$$

$$b = 1$$

Unique soln

$$(m, 7, -4)$$

is a linear combination

$$(-2, 2, 1) \text{ and } (2, 1, -2)$$

$$\begin{bmatrix} m \\ 7 \\ -4 \end{bmatrix} = a \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} m \\ 7 \\ -4 \end{bmatrix} = \begin{bmatrix} -2a + 2b \\ 2a + b \\ a - 2b \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -2 & 2 & m \\ 2 & 1 & 7 \\ 1 & -2 & -4 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cc|c} 1 & -2 & -4 \\ 2 & 1 & 7 \\ -2 & 2 & m \end{array} \right] \begin{array}{l} R_2 = -2R_1 + R_2 \\ R_3 = 2R_1 + R_3 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & -2 & -4 \\ 0 & 5 & 15 \\ 0 & -2 & m-8 \end{array} \right] \xrightarrow{R_2 = \frac{1}{5}R_2} \left[\begin{array}{cc|c} 1 & -2 & -4 \\ 0 & 1 & 3 \\ 0 & -2 & m-8 \end{array} \right] \begin{array}{l} R_1 = 2R_2 + R_1 \\ R_3 = 2R_2 + R_3 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & m-2 \end{array} \right] \quad \therefore m-2=0$$

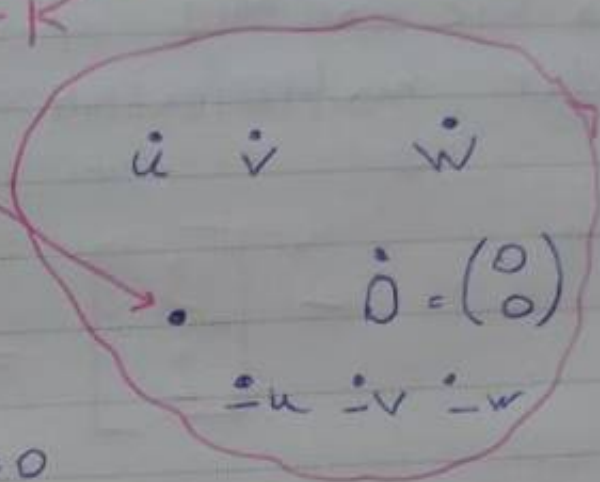
$$\boxed{m=2}$$

The set $V = \mathbb{R}^2$ with the standard operations of addition and scalar multiplication is a vector space

$$u = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + v = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix} \quad V = \mathbb{R}^2$$

$$w = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

$$0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



$$u + 0 = u$$

$$u + (-u) = 0$$

$$-u = \begin{pmatrix} -x_1 \\ -x_2 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} -x_1 \\ -x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$c \in \mathbb{R}$$

$$cu + cv = c(u + v)$$

$$\mathbb{R}^2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Determine whether $V = \{(\underline{x}, \underline{x}) : x \text{ is a real number}\}$ together with the standard operations, is a vector space. If it's not, identify at least one of the ten vector space axioms that fails.

$$cu = (cu_1, cu_2) \in V$$

$$1u = (u_1, u_1) \in V$$

$$d(cu) = d(cu_1, cu_2)$$

$$= (dcu_1, dcu_2) \in V$$

$$dc(u) = (dcu_1, dc u_2) \in V$$

V is a vector space

$$V = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$u = (u_1, u_2)$$

$$-u = (-u_1, -u_2) \quad \vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\vec{v} = (v_1, v_2)$$

$$R_3 = -2R_2 + R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right]$$

$$R_3 = -R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$R_1 = 2R_3 + R_1$$

$$R_2 = 4R_3 + R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$\therefore C_1 = C_2 = C_3 = 0$
(Trivial solution)

$\therefore S$ is Linearly
Independent

□